



Advanced oxidation processes for the removal of organophosphorus pesticides in aqueous matrices: A systematic review and meta-analysis



Mohammad Malakootian^{a,b}, Armita Shahesmaeili^c, Maryam Faraji^{a,b}, Hoda Amiri^{a,b,*}, Susana Silva Martinez^{d,**}

^a Environmental Health Engineering Research Center, Kerman University of Medical Sciences, Kerman, Iran

^b Department of Environmental Health, School of Public Health, Kerman University of Medical Sciences, Kerman, Iran

^c HIV/STI Surveillance Research Center, and WHO Collaborating Center for HIV Surveillance, Institute for Futures Studies in Health Kerman University of Medical Sciences, Kerman, Iran

^d Centro de Investigación en Ingeniería y Ciencias Aplicadas, Av. Universidad 1001, Col. Chamilpa, Cuernavaca, Morelos, Mexico

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ABSTRACT

The advanced oxidation processes (AOPs), as an alternative technology to eliminate pesticides from aqueous environments, consist of several groups of technologies that have been used with high efficiency in the treatment of water and wastewater in recent decades. A systematic review of the scientific literature to evaluate the most common advanced oxidation processes (AOPs) for the removal of organophosphorus pesticides in aqueous matrices is addressed in this study. Meta-analysis is also performed to provide a precise and robust summary estimate after a systematic and rigorous integration of the available evidence. In the current study, 9 sub-groups of AOPs were reviewed, such as electrochemical, UV/H₂O₂ photolysis, photocatalysis, Fenton-type, plasma, gamma irradiation, sulfate-based catalyst, sonolysis and ozonation technology for organophosphorus pesticides degradation. The random effects model was used to estimate the pooled measurements and 95 % confidence intervals (95 % CI). In total, six studies were included in this review. All studies, except one, used the photocatalytic process as AOP. The average pooled percentage of AOP for pesticide degradation was 66.8 (95 % CI: 58.1–75.6). In addition, the most studied pesticides are chlorpyrifos and diazinon which, according to the results of the meta-analysis, the photocatalytic process has the highest efficiency of diazinon elimination with an average percentage of 79.2 (95 % CI: 76.8–81.5).

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* Corresponding author at: Department of Environmental Health, School of Public Health, Kerman University of Medical Sciences, Kerman, Iran.

** Corresponding author at: Centro de Investigación en Ingeniería y Ciencias Aplicadas, Av. Universidad 1001, Col. Chamilpa, Cuernavaca, Morelos, México.
E-mail addresses: hoda.amiri@gmail.com (H. Amiri), ssilva@uaem.mx (S. Silva Martinez).

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1. Introduction

Pesticides are a widespread group of chemicals used to improve agricultural production and home care. However, these compounds can be classified into a category of persistent organic pollutants that remain in water for a long time, accumulate in sediments or bioaccumulate in biota, are transferred over long distances or have toxic effects to the environment, human health and living organisms even in low concentrations. Said classification is based on the solubility and Log Kow (a measure of their hydrophobicity) of these compounds. People are exposed to pesticides through consumption of contaminated drinks, food and other environmental media (Katsikantami et al., 2019).

Organophosphorus pesticides, the most common pesticides, account for 34 % of world sales which according to the published papers chlorpyrifos, diazinon, malathion, parathion and phorate are the most prevalent in the environment (Amiri et al., 2018; Katsikantami et al., 2019; Khedr et al., 2019; Naddafi et al., 2018). To reduce the potential health risks, the AOPs are an alternative technology to remove pesticides from the environment. They are non-selective processes for the removal of organic pollutants, by producing oxidizing radical species (e.g., $H^{\bullet}$, $O_2^{\bullet-}$, $O_3^{\bullet-}$, $OH^{\bullet}$, etc.) capable of degrading and mineralizing contaminants in CO_2 , H_2O and inorganics (Jaafarzadeh et al., 2014; Mahvi et al., 2011; Yaghmaeian et al., 2016).

Although, the literature shows some priority in the tendency to use AOPs for the degradation of organophosphorus pesticides in recent years, it seems that processes such as photocatalysis (Amiri et al., 2018; Naddafi et al., 2018; Toolabi et al., 2019), Fenton and electro-Fenton (Dominguez et al., 2018), radiation (UV, gamma and ozone) (Bustos et al., 2019; Cruz-Alcalde et al., 2018; Khedr et al., 2019) and persulfate based AOP (Liu et al., 2019) are the most recently used technology for the elimination of organophosphorus pesticides.

This review, although by no means exhaustive, presents a systematic review of the literature to evaluate the most common advanced oxidation processes used for the removal of organophosphorus pesticides in aqueous media. In addition, a meta-analysis is carried out to provide a precise and robust summary estimate after a systematic and rigorous integration of the available evidence. To the best of our knowledge, we are the first to use meta-analysis to investigate this subject.

2. Chemical structure of the organophosphorus compounds

Organophosphorus compounds (OP) contain a pentavalent phosphorus atom with a double bond to either oxygen ($P=O$) or Sulphur ($P=S$). The chemical structures of OP and common OP are shown in Fig. 1 (Kang et al., 1995). X is the leaving group with structures ranging from a single halogen atom to a complex substituted aryl ring (Ballantyne and Marrs, 2017) and is the principal metabolite for a specific identification (Kazemi et al., 2012).

3. Methodology

3.1. Data sources and search strategy

A systematic search of the peer-reviewed literature was carried out in the Web of science and Science direct databases as scientific articles that evaluated the oxidation processes for the treatment of aqueous solution contaminated with pesticides. In the systematic review, we searched articles that were published until January 27, 2019. The search included the keywords [(Advance oxidation process) OR (AOP)] AND (organophosphorus pesticide) AND [(water) OR (wastewater) OR (aqueous) OR (agriculture runoff)] AND [(treatment) OR (removal) OR (decomposition) OR (degradation) OR (mineralization)]. The literature search was limited to peer-reviewed publications written in English between 2010 until January 2019. After this stage, we considered a set of inclusion and exclusion criteria, which are described below:

The study inclusion criteria apply to each publication, which consists of scope (Step 1), study quality (Step 2), and data availability (Step 3). For Step 1, two independent screeners first evaluated the titles and abstracts of the retrieved articles to assess whether the paper included organophosphorus pesticides removal using AOPs in aqueous solution. In addition, the full text of the papers whose abstracts passed the first screening step to confirm that the document contained an experimental study to evaluate the efficiency of the organophosphorus pesticide oxidation process. We excluded books, presentations, review papers, and letter to the editor about AOPs for the removal of organophosphorus pesticides and other environmental matrices such as soil and air. Also, papers about the development of detection methods of organophosphorus pesticides in different media were excluded. Information on each paper was extracted, such as the first author, year of publication, type of pesticide, type of oxidation process, response factor for the evaluation of the efficiency of the AOP and optimum pH. For Step 2, the quality of a study was evaluated independently by two scientific reviewers. The studies have passed the criteria of clarity. Publications in which their study and associated methodologies were not sufficiently documented to investigate the quality of the study were not included. After a publication passed both scope and quality criteria, the availability of the data (step 3) was analyzed. For this selection step, a requirement was that the publications include experimental design through the use of different software, and the available answers imply the mean and standard deviation of data.

3.2. Statistical analysis

To calculate the combined percentage of AOP efficiency for the removal of organophosphorus pesticides, studies designed by response surface methodology (RSM) entered the meta-analysis. In studies with multiple executions, the average removal percentage for all executions was calculated. The pooled estimate was calculated using the `metan` command in the Stata software Version 14.

Phosphorus group	General structure	Common pesticides
Phosphate	$\begin{array}{c} \text{O} \\ \parallel \\ \text{R}-\text{O}-\text{P}-\text{O}-\text{X} \\ \diagup \\ \text{R}-\text{O} \end{array}$	Chlorfenvinphos, Dichlorvos
Phosphonate	$\begin{array}{c} \text{O} \\ \parallel \\ \text{R}-\text{O}-\text{P}-\text{X} \\ \diagup \\ \text{R}-\text{O} \end{array}$	Butonate, Trichlorfon
Phosphonothioate	$\begin{array}{c} \text{S} \\ \parallel \\ \text{R}-\text{O}-\text{P}-\text{O}-\text{X} \\ \diagup \\ \text{R} \end{array}$	Trichlormat, Cyanofenphos
O-alkyl phosphorothioate	$\begin{array}{c} \text{S} \\ \parallel \\ \text{R}-\text{O}-\text{P}-\text{O}-\text{X} \\ \diagup \\ \text{R}-\text{O} \end{array}$	Demeton-S-methyl, Vamidithion
	$\begin{array}{c} \text{O} \\ \parallel \\ \text{R}-\text{O}-\text{P}-\text{S}-\text{X} \\ \diagup \\ \text{R}-\text{O} \end{array}$	Bromophos, Chlorpyrifos, Diazinon
Phosphorodithioate	$\begin{array}{c} \text{S} \\ \parallel \\ \text{R}-\text{O}-\text{P}-\text{S}-\text{X} \\ \diagup \\ \text{R}-\text{O} \end{array}$	Dimethoate, Disulfoton, Malathion
S-alkyl phosphorothioate	$\begin{array}{c} \text{O} \\ \parallel \\ \text{R}-\text{S}-\text{P}-\text{O}-\text{X} \\ \diagup \\ \text{R}-\text{O} \end{array}$	Profenofos, Trifenofos
S-alkyl phosphorodithioate	$\begin{array}{c} \text{S} \\ \parallel \\ \text{R}-\text{S}-\text{P}-\text{O}-\text{X} \\ \diagup \\ \text{R}-\text{O} \end{array}$	Prothiofos, Sulprofos
Phosphoramidate	$\begin{array}{c} \text{O} \\ \parallel \\ \text{R}-\text{O}-\text{P}-\text{N} \begin{array}{l} \diagup \text{R} \\ \diagdown \text{R} \end{array} \\ \diagup \\ \text{R}-\text{O} \end{array}$	Cruformate, Fenamifos
Phosphorotriamidate	$\begin{array}{c} \text{O} \\ \parallel \\ \text{R} \diagup \text{N}-\text{P}-\text{N} \diagdown \text{R} \\ \\ \text{N} \\ \diagup \text{R} \diagdown \text{R} \end{array}$	Triamifos
Phosphorothioamidate	$\begin{array}{c} \text{O} \\ \parallel \\ \text{R}-\text{O}-\text{P}-\text{N} \begin{array}{l} \diagup \text{R} \\ \diagdown \text{R} \end{array} \\ \\ \text{S-alkyl} \end{array}$	Methamidofos
	$\begin{array}{c} \text{O} \\ \parallel \\ \text{R}-\text{O}-\text{P}-\text{N} \begin{array}{l} \diagup \text{R} \\ \diagdown \text{R} \end{array} \\ \diagup \\ \text{R}-\text{O} \end{array}$	Isofenfos

Fig. 1. Structures of organophosphorus compounds commonly used as pesticides.

The random effects model was used to estimate the pooled measurements and 95 % confidence intervals (95 % CI). The subgroup analysis was performed only in groups with at least two studies. The forest digrammas were used to show the results of the meta-analysis. All analyzes were performed with STATA version 15.0 (STATA Corp, College Station, Texas, USA).

4. Results

4.1. Study selection

In recent years, a growing number of publications have been devoted to the study of the treatment of pesticides by oxida-

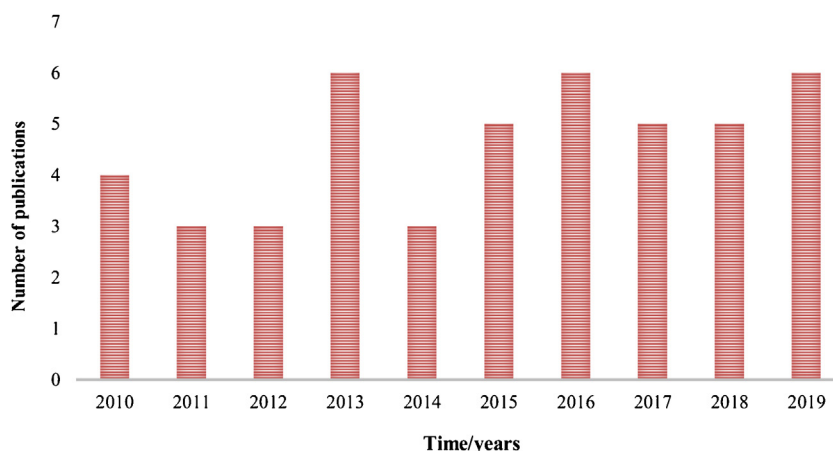


Fig. 2. Number of publications during the last decade in the Web of science database (46).

tive techniques and AOPs (Fig. 2). The classification of published papers according to the AOPs used, the aqueous matrices and the organophosphorus pesticides are shown in Fig. 3. The study selection considered four stages, such as identification, screening, eligibility and inclusion of the documents. Thus, in the first step of the selection procedure, 335 records were identified as shown in the PRISMA flow-diagram of Fig. 4. Sixty duplicates and 221 records were removed because they did not meet the inclusion criteria (they were not published in English, were review articles, encyclopedia, book chapters, conference summary, mini reviews, news, short communications and others), so the eligibility of 54 studies was evaluated. In the third step (identification of redundancies among the full text papers that will be evaluated to determine their eligibility), 8 text papers that will be evaluated to determine their eligibility, 8 studies were excluded and 46 were included in the qualitative synthesis. Therefore, 6 studies were identified for inclusion in the meta-analysis after the selection process was completed.

4.2. Degradation of organophosphorus pesticides by AOPs

Hydroxyl radicals ($\text{OH}^\bullet$) are the most powerful oxidants (oxidation potential of 2.72 V) and are responsible for the degradation of most organic pollutants in water. The degradation of aqueous organic compounds mainly depends on the amount of hydroxyl radicals produced through different AOPs. Many other oxidant species are produced in the AOPs in addition to $\text{OH}^\bullet$ radicals, such as superoxide radical anions ($\text{O}_2^{\bullet-}$), hydroperoxyl radical ($\text{HO}_2^\bullet$), hydrogen peroxide (H_2O_2), solvated electrons (e^-), hydrogen radicals ($\text{H}^\bullet$), amongst others.

According to the literature, the possible degradation mechanism of organophosphorus pesticides involves the formation of hydroxylated intermediates (a mixture of non-retained species of high polarity such as phosphate derivatives) as the first step in the AOPs, followed by the formation of aliphatic compounds after bond-breaking reactions (Dzyadevych and Chovelon, 2002; Guivarch et al., 2003; Herrmann et al., 1999). The products of ring opening, and side chain breaking might be alcohols, aldehydes or ketones, which will be oxidized to carboxylic acids and mineralized to CO_2 and H_2O (Guivarch et al., 2003; Herrmann et al., 1999).

This section reports the findings of the degradation of organophosphorus pesticides using different AOPs, such as electrochemical, UV/ H_2O_2 photolysis, photocatalysis, Fenton-type, plasma, gamma irradiation, sulfate-based catalyst, sonolysis and ozonation technology.

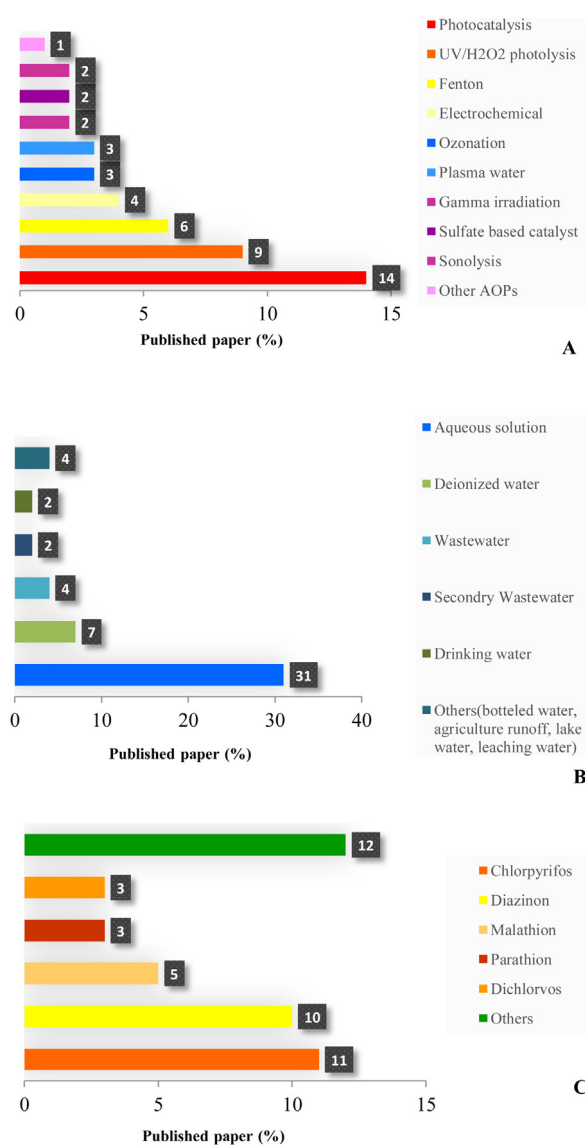


Fig. 3. Classification of published paper(percent) based on the AOPs studied (A), water matrices (B) and organophosphorus pesticides (C).

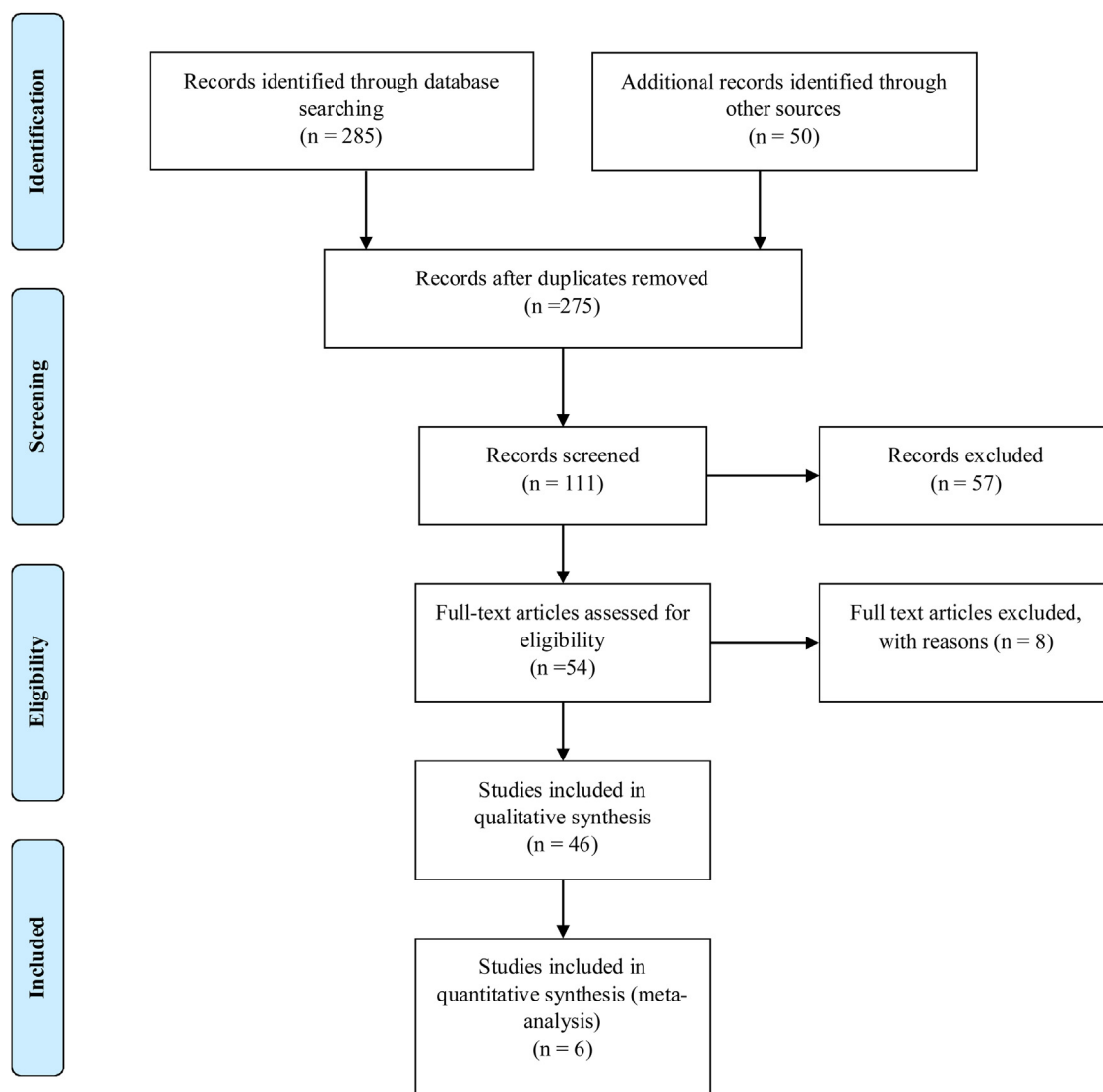


Fig. 4. The PRISMA flowchart of the search and selection of papers.

4.2.1. Electrochemical processes

The electrochemical methods produce hydroxyl radicals using direct or indirect electrochemical oxidation technologies. These technologies include electrochemical, sonoelectrochemical and photoelectrochemical processes and involve surface and bulk oxidation processes. Different electrochemical processes have been used to study the degradation of pesticides such as chlorpyrifos, glyphosate, karbofos (malathion), bazudin (diazinon), chlorophos, metaphos and thiamethoxam in aqueous solutions with a wide range of initial concentrations at laboratory scale. Table 1 includes the experimental conditions, the kinetics and the degree of degradation of the pesticides mentioned above. Samet and collaborators studied the anodic oxidation of chlorpyrifos at Nb/PbO₂ electrode using bulk electrolysis (Samet et al., 2010). These investigators observed an increase in the chemical oxygen demand (COD) removal by increasing the apparent current density from 10 to 50 mA cm⁻² because more OH[•] were generated; however, the instantaneous current efficiency (ICE) dropped to zero with prolonged electrolysis, since a large part of these radicals was wasted, which hampered the complete mineralization. A 76 % of COD removal was achieved at 10 h of electrolysis under the best experimental conditions. In addition, they observed that even the ICE fell when the oxidation of chlorpyrifos took place in diluted solutions

(Samet et al., 2010). The degradation of chlorpyrifos was effectively improved by the combination of electrolysis with sonolysis (US-EC) using common stainless-steel electrodes. Ren and co-workers reported a synergistic factor of 37 % compared to the treatments of US and EC (Ren et al., 2019). In addition, other organophosphorus compounds, such as glyphosate, karbofos (malathion), bazudin (diazinon), chlorophos and metaphos were also evaluated by the electrochemical process (Kukurina et al., 2014). These authors reported that the commercial pesticides have lower electrolysis efficiency compare to the analytical grade purity. This was attributed to the presence of bulkier materials (e.g. dyestuffs, surface active agents and talc) in commercial pesticide products. In general, according to COD data and phosphate ion accumulation, it was estimated that pesticides were completely destroyed following pseudo first order kinetics (Kukurina et al., 2014). Finally, thiamethoxam (TMX) degradation over a boron-doped diamond (BDD) anode was investigated by Lebik-Elhadi and co-workers. They studied the effect of different operating parameters, such as initial pollutant concentration, current intensity, presence of humic acid, bicarbonate and chlorine on TMX degradation kinetics. In addition, the degradation and mineralization of TMX in a real water matrix was observed. According to their results, the C/C₀ ratio in different aqueous matrices followed the order: wastew-

Table 1

Summary of the optimal conditions of the electrochemical processes for the removal/degradation of pesticides and kinetic data.

Pesticide	Initial concentration and Matrix-Scale	AOP features	Kinetic data	Reference
Chlorpyrifos	COD = 450 mg O ₂ L ⁻¹ in aqueous solution	Niobium (Nb)/PbO ₂ electrodes, pH = 2, T = 70 °C, 50 mAcm ⁻²	pseudo second-order kinetics, Removal efficiency(10 h) ~76 % COD	(Samet et al., 2010)
Chlorpyrifos	900 mg L ⁻¹ in aqueous solution	Electrode in the EC = stainless-steel combine with US, pH = Neutral, 15–35 °C	Removal efficiency EC ~72.8 % < US-EC ~ 93.3 %	(Ren et al., 2019)
Glyphosate, Karbofos (Malathion), Bazudin (Diazinon), Chlorophos, Metaphos	Chlorophos = 3.7 g L ⁻¹	Electrode in the EC = Lead	Removal efficiency	(Kukurina et al., 2014)
	Metaphos = 9.4 g L ⁻¹ in polluted aqueous sulfuric acidic solutions	current density = 1.2 A/cm ²	(240 min), Chlorophos = 100% Metaphos = 99.57 %	
Thiamethoxam	1–10 mg L ⁻¹ in pure water, secondary treated wastewater, bottled water	Electrochemical, boron doped diamond anode pH = 3–11	pseudo-first order kinetic	(Lebik-Elhadi et al., 2018)

ater < bottled water < wastewater diluted by 50 % with ultrapure water < ultrapure water (Lebik-Elhadi et al., 2018).

4.2.2. UV/H₂O₂ photolysis process

UV/H₂O₂ photolysis is a process in which organophosphorus compounds absorb photons and the energy released drives oxidation processes induced by light. Some organic compounds can be reduced by photolysis alone. The reduction of organophosphorus compounds, such as those reported in Table 2, is aided by the addition of hydrogen peroxide to generate hydroxyl radicals in an advanced oxidation process.

Table 2 reports the degradation of parathion, chlorpyrifos (CPF), fensulfothion, diazinon and malathion by H₂O₂/UV photolysis in aqueous solutions. Wu and co-workers (Wu and Linden, 2010) examined the influence of anions such as bicarbonate and carbonate on the hydroxyl radical production in the UV/H₂O₂ process during the degradation of parathion and chlorpyrifos under UV light irradiation by low-pressure Hg lamps emitting at 254 nm. These authors observed that the addition of hydrogen peroxide increased the UV degradation rates of both pesticides while the presence of bicarbonate and carbonate ions reduced the pesticide degradation rates via scavenging of hydroxyl radical (Wu and Linden, 2010). In addition to a high degree of oxidative degradation of fensulfothion by .OH (~83 % TOC reduction), the use of a high concentration of H₂O₂ (5×10^{-2} mol dm⁻³) was observed during 80 min of irradiation (Sunil Paul et al., 2013). Nearly complete (~90 %) transformation of fensulfothion was detected within 4 min of irradiation even though very little total organic carbon reduction was obtained at this time scale (Sunil Paul et al., 2013). The formation of intermittent oxidation by-products as a cause of lack of complete degradation of the organic compound is a major disadvantage of AOPs.

Several disinfection by-products (DBP) of chlorinated diazinon solution were formed after illumination with UV and UV/H₂O₂ peroxidation. Such DBP were monochloroacetic acid (MCAA), dichloroacetic acid (DCAA), trichloroacetic acid (TCAA), chloroform (TCM), dichloroacetonitrile (DCAN) and 1,1,1-trichloroacetone (1,1,1-TCP) (Li et al., 2015). The solution pH and H₂O₂ dose affected UV-oxidation byproducts speciation which altered DBPs

formation. The DBP formation increased significantly in the treated diazinon solutions with UV irradiation (Li et al., 2015). In addition, four main UV oxidation byproducts [2-isopropyl-6-methyl-4-pyrimidinol (IMP), O-analog diazinon (diazoxon), diethyl thiophosphate (DETP) and diethyl phosphate (DEP)] were examined to identify their relative contributions to DBPs formation. The increase in total DBPs formation of the treated diazinon solutions was attributed mainly to its oxidation product IMP and its secondary oxidation products, while DETP and DEP had little effect. Moreover, its oxidation fragment diazoxon intensified the formation of MCAA, DCAA and TCAA under UV/H₂O₂ pre-oxidation condition. Malathion was another pesticide treated with the H₂O₂/UV process (Chenna et al., 2016). It was reported that the photodegradation of malathion was more effective for higher values of pH and temperature and its degradation kinetics was a first order reaction.

The ecotoxicity and phytotoxicity of the aqueous solutions treated before and after the oxidative process was a considerable factor in selecting the AOP process (Utzig et al., 2019). In this regard, several scientists preferred the UV/H₂O₂ and UVC radiation processes for the oxidation of CPF, and the efficiency of the treatment was evaluated considering the acute toxicity of the *Aedes aegypti* larvae and the seed germination/ root elongation test with *Lactuca sativa* seeds. The results showed that, although both processes were efficient, showing a reduction of over 97 % of initial CP after 20 and 60 min of UV/H₂O₂ and UVC radiation, respectively, toxicity could increase and lead to larvae mortality (> 90 % of organisms) and inhibition effects on seed root growth (Utzig et al., 2019).

4.2.3. Photocatalytic treatment

The elimination of organophosphorus pesticides in the photocatalytic process takes place on the catalyst surface upon chemical reactions produced between the oxidant species (valence band holes $-h^+$, $\text{OH}^\bullet$, $\text{O}_2^{\bullet-}$, H_2O_2 , etc.) and the organophosphorus pesticides in the wastewater. Such oxidizing species are generated in the process of catalyst activation with adequate light irradiation. In general, the complexity of the photocatalytic process leads to the consideration of several operational factors, such as pH, concentration of catalyst or oxidant, light intensity and substrate

Table 2Summary of the UV/H₂O₂ photolysis process for the removal/degradation of pesticides.

Pesticide	Initial concentration and Matrix-Scale	AOP features	Kinetic data	Reference
Parathion (Para), Chlorpyrifos (CPF)	Para = 5 μ M CPF = 3–5 μ M in aqueous solution	UV/H ₂ O ₂ by low-pressure Hg lamps pH = 7	second-order rate constants	(Wu and Linden, 2010)
Fensulfothion	1 $\times$ 10–4 mol dm ⁻³ in aqueous solution	Pulse radiolysis + H ₂ O ₂ /UV photolysis, pH ~ Neutral	~90 % Fensulfothion removal (4 min) ~83 % TOC removal (80 min)	(Sunil Paul et al., 2013)
Diazinon	5 mg L ⁻¹ and 16.45 μ M in aqueous solution	Photochemical reactor with a low-pressure mercury UV lamp pH = 4,7,10	pseudo first-order degradation	(Li et al., 2015)
Malathion	10–100 mg L ⁻¹ in demineralized water	Static reactor with a low pressure mercury vapor lamp pH = 3–12	1.0 % removal of Malathion under optimal condition, first order reaction	(Chenna et al., 2016)
Chlorpyrifos	200 μ g L ⁻¹ in aqueous solution	A borosilicate reactor with a high pressure mercury vapor lamp pH = Neutral	UV/H ₂ O ₂ = 97% UVC = 97 %	(Utzig et al., 2019)
Phorate	12 mgL ⁻¹ in aqueous solution	TiO ₂ + two UV-365 nm lamps (15 W), TiO ₂ dosage = 0.1 g L ⁻¹ pH = 8	Phorate removal (60 min) = 99 % pseudo-first-order kinetics	(Wu et al., 2010)
Dimethoate	10 ppm In aqueous solution	UV irradiation + TiO ₂ /polymer films TiO ₂ dosage = 4 g L ⁻¹ pH = 4.62	Dimethoate removal (180 min) ~ 100% TOC mineralization (300 min) ~ 50 %	(Priya et al., 2011)
6chloronicotinic acid (6CNA)	51.2 $\pm$ 0.2 mg L ⁻¹ Using double deionized water	TiO ₂ +UVA polychromatic fluorescent lamps pH = 4.23 $\pm$ 0.01	first-order kinetics TOC mineralization (120 min) ~ 46 $\pm$ 7 %	(Žabar et al., 2011)
Fenamiphos	0.1 mg L ⁻¹ in leaching water	zinc oxide (ZnO)+ TiO ₂ , tungsten (VI) oxide (WO ₃), and tin (IV) oxide (SnO ₂) + Neutral sunlight, solar pilot plant (parabolic collector (CPC)), pH 8.2 pH = 8.2	ZnO $\geq$ TiO ₂ P25 Degussa (99.9%) > TiO ₂ Anatase (99.7%) $\geq$ TiO ₂ anatase:rutile (1:3) (99.4%) < WO ₃ (68.0%) < SnO ₂ (59.1%) < Photolysis (7.3%)	(Fenoll et al., 2012)

concentration (e.g. interference of anions) that influence the oxidation of organic compounds in order to find optimal operating conditions (Flint and Suslick, 1991). The pH plays an important role in the photocatalytic oxidation of organic contaminants. For instance, it modifies the surface charge of the semiconductor, affects the interfacial electron transfer, and alters the photoredox process and the hydroxyl radical generation mechanism. The results of the phorate photocatalysis showed that the rate of degradation decreased with a decrease in pH, and was much higher in alkaline conditions (Wu et al., 2010). In addition, the pH on a photocatalytic reaction can be an interference of the surface charge of the nanocatalyst and anionic or cationic pollutant. In non-ionic pesticides, the presence of large amounts of OH⁻ ions in alkaline solutions may be a major factor in improving the rate of degradation. By contrast, some pesticides (e.g. dimethoate, 6chloronicotinic acid, fenamiphos, imazalil, monocrotophos and chlorpyrifos, methyl parathion and parathion, acetamiprid and diazinon) are more efficiently degraded in acidic to near neutral condition (Amalraj and Pius, 2015; Amiri et al., 2018; Fenoll et al., 2012; Flint and Suslick, 1991; Hazime et al., 2013; Hoffmann et al., 1996; Jonidi-Jafari et al., 2015; Naddafi et al., 2018; Priya et al., 2011; Sharma et al., 2016; Shirzad-Siboni et al., 2017; Suslick, 1990; Tabasideh et al., 2017; Žabar et al., 2011, 2016). One reason may be the point of zero charge (pzc) of the commercial TiO₂ catalyst which is around 6.0, that below and above this pzc, the catalyst

surface is positively or negatively charged, respectively. Another reason is the scavenging of photo generated holes, that yields highly oxidative •OH species through the interaction with H⁺ ions, causing a high formation of hydroxyl ions. Moreira and collaborators treated a mix of several pesticides in wastewater resulting from the phytopharmaceutical plastic containers washing, after the biological treatment, using the TiO₂/UV pilot scale system. These authors reported a degradation rate of more than 90 % for eighteen pesticides, in previously acidified solutions, where thirteen of them were removed at levels below respective limits of quantification. A minimum removal of 70 % was achieved for the rest of the pesticides, with the exception of the pesticide terbutylazine-desethyl, which registered a significant increase (11 times) (Moreira et al., 2012). Rani and Shanker (Rani and Shanker, 2018) reported that neutral conditions produced the maximum degradation of three pesticides (α -hexachloro cyclohexane (α -HCH), malathion and chlorpyrifos) by the Prussian blue (Iron hexacyanoferrate; FeHCF) nanorods catalyst under the sunlight. These authors commented that FeHCF catalyst has a low band gap (1.145 eV), which makes it an adequate and a safe photocatalyst via green route for the oxidation of several organic compounds (Rani and Shanker, 2018). With respect to catalyst loading, it has been observed that increasing the number of catalyst particles increases the surface area of the catalyst, the number of photons, the creation of a higher number of active species and the number of contaminating molecules

absorbed. Conversely, when the catalyst dose and the initial contaminant concentration were overdosed, the overall photocatalyst performance was reduced by lower light penetration, greater light scattering, and available reaction sites (Amalraj and Pius, 2015; Amiri et al., 2018; Flint and Suslick, 1991; Hazime et al., 2013; Hoffmann et al., 1996; Jonidi-Jafari et al., 2015; Mirmasoomi et al., 2017; Priya et al., 2011; Sharma et al., 2016; Shirzad-Siboni et al., 2017; Suslick, 1990; Tabasideh et al., 2017; Wu et al., 2010). In addition, high-strength inorganic ions, present in aqueous environments and agriculture runoff, can compete for active sites on the surface of the nanocatalyst and, therefore, can decrease the photo mineralization/degradation rate of pesticides. Many studies have reported the effects of inorganic anions and organic species in the inhibition of the photocatalytic properties of the catalysts. Such as Cl^- and NO_3^- (Wu et al., 2010), total nitrogen ($1.8 \pm 0.16 \text{ mg L}^{-1}$) and total phosphorous ($0.34 \pm 0.01 \text{ mg L}^{-1}$) (Flint and Suslick, 1991), carbonate (CO_3^{2-} , hardness 86 mg L^{-1}) and sulfate (SO_4^{2-} concentration 168.8 mg L^{-1}) (Sharma et al., 2016), citric acid, humic acid, oxalate, EDTA, phenol, folic acid, citric acid, each organic species at 20 mg L^{-1} (Jonidi-Jafari et al., 2015; Shirzad-Siboni et al., 2017)). The inhibition of the photocatalytic properties in the presence of ions is often explained by the scavenging of hydroxyl radicals due to the reaction of positive holes and hydroxyl radical with ions.

The photocatalytic process under UV illumination has been widely used for the treatment of different organophosphorus pesticides. Among these pesticides are phorate (Wu et al., 2010), dimethoate (Priya et al., 2011), 6-chloronicotinic acid (Žabar et al., 2011), fenamiphos (Fenoll et al., 2012), smetolachlor, 2,4-d, mcpa, imidacloprid, alachlor, terbuthylazine, isoproturon, bentazone, tebuconazole, atrazine, linuron, metobromuron, dimethoate, diuron, metribuzin, metalaxy I, chlorotoluron, simazine, terbuthylazine-desethyl, benalaxyl, carbaryl, carbofuran, cymoxanil, deisopropylatrazine, desethylatrazine, methidathion, propanil, propazine, propylenethiourea, pyrimethanil, triclopyr (Moreira et al., 2012), imazali (Hazime et al., 2013), monocrotophos and chlorpyrifos (Amalraj and Pius, 2015; Amiri et al., 2018; Flint and Suslick, 1991; Rani and Shanker, 2018), α -hexachloro cyclohexane (α -hch) and malathion (Rani and Shanker, 2018), 3,5,6-trichloro-2-pyridinol (Žabar et al., 2016), methyl parathion, parathion (Sharma et al., 2016), acetamiprid (Hoffmann et al., 1996) and diazinon (Jonidi-Jafari et al., 2015; Mirmasoomi et al., 2017; Shirzad-Siboni et al., 2017; Suslick, 1990; Tabasideh et al., 2017), as shown in Table 3.

4.2.4. Homogeneous and heterogeneous Fenton-type processes

The Fenton process, and Fenton-type processes, is based on the reaction between iron ions and hydrogen peroxide to produce hydroxyl radicals at ambient temperature. Among AOPs, the Fenton-type method can be an effective method for the removal of contaminants. Several advantages of the Fenton process have been identified, such as safe and environmentally-benign nature of reagents, relatively simple operating principles, short reaction time, and absence of mass transfer limitations (Kang and Hoffmann, 1998; Suslick, 1990). Both the homogeneous and the heterogeneous Fenton processes (operating at or near neutral pH) have been used successfully as pre-treatment and second treatment for the degradation of pesticide. Table 4 summarizes the findings for the degradation of pesticides using several Fenton-type processes. Based on the efficiency removal of organic compounds, the pH is a major factor in the Fenton process because of iron and hydrogen peroxide speciation factors. By increasing the pH, the activity of the Fenton reagent is reduced due to the presence of relatively inactive iron oxo hydroxides and the formation of ferric hydroxide precipitate. This phenomenon reduces the generation of hydroxyl radicals, since fewer free iron ions are available to catalyze the H_2O_2 . A pH = 3 was reported as the optimal pH of the solution for the degrada-

tion of the several pesticides, such as malathion (Zhang and Pagilla, 2010), methyl parathion (Patil and Gogate, 2012), chlorpyrifos, cypermethrin and chlorothalonil (Affam et al., 2014), chlorfenvinphos (CFVP) (Oliveira et al., 2014), triazophos (Gogate and Patil, 2015) and diazinon (Wang and Shih, 2015).

The amount of Fe(II)/Fe(III) and hydrogen peroxide (molar ratio of $\text{Fe}^{\text{III/II}}/\text{H}_2\text{O}_2$) affects the production rate and concentration of the highly active intermediates, such as OH $\cdot$ radicals. High concentrations of H_2O_2 leads to the radical scavenging effect, the self-decomposition of H_2O_2 in O_2 and H_2O , and the additional costs of operation; therefore, the optimal concentration of H_2O_2 must be determined in the Fenton-based processes (Malakootian and Moridi, 2017). Different optimal values of the $\text{Fe}^{\text{III/II}}/\text{H}_2\text{O}_2$ molar ratio has been reported depending on the chemistry of the pesticides: $\text{Fe(II):H}_2\text{O}_2$ ratio of 1:40 for the degradation of malathion (Zhang and Pagilla, 2010), 4:1 for methyl parathion (Patil and Gogate, 2012), 1:25 for chlorpyrifos, cypermethrin and chlorothalonil (Affam et al., 2014), 1:32.6 for chlorfenvinphos (Oliveira et al., 2014), 1:4 for triazophos (Gogate and Patil, 2015) and, 1:7.5 for Diazinon removal (Wang and Shih, 2015). Therefore, the H_2O_2 load requirement is different for each pollutant under study and, therefore, it is important to conduct studies for specific contaminants. With regarding to iron concentration, it has been reported that at high iron concentration malathion degradation decreased with the addition of iron salt (Affam et al., 2014; Oliveira et al., 2014; Patil and Gogate, 2012; Wang and Shih, 2015; Zhang and Pagilla, 2010). The degradation of triazophos was enhanced by the addition of Fenton reagent under sonication due to the intensified production of hydroxyl radicals as a result of the conversion of Fe^{III} to Fe^{II} under the cavitating conditions (Gogate and Patil, 2015). It has been shown that increasing the initial concentration of contaminants reduces the removal efficiency in the Fenton process. This occurs because high concentrations of contaminants require higher concentrations of hydroxyl radicals for effective removal, while the formation of these reactive species is constant. In addition, in the heterogeneous process of photoFenton, the number of active sites, where the radical reactions take place and also the hydrogen peroxide absorbs light, is limited when the concentration of contaminant is too high. The result of degradation of malathion (Zhang and Pagilla, 2010), methyl parathion (Patil and Gogate, 2012), chlorpyrifos, cypermethrin and chlorothalonil (Affam et al., 2014) showed that the reaction rate of the high concentration was slower than that of the initial low pesticide concentration.

4.2.5. Plasma technology

Pulse electrical discharge plasma is an ionized gas consisting of electrons, free radicals, ions and neutrals. Plasma-based water treatment technology (PWT) involves the generation of hydroxyl radicals *in situ*, in addition to other reactive species ($\cdot\text{OH}$, $\text{O}^\cdot$, $\text{H}^\cdot$, $\text{HO}_2^\cdot$, O_2^- , O_3 , H_2O_2 , H_2), and from the water itself. Therefore, unlike most AOPs and conventional processes, few or no chemical additives are required. PWT can be performed as thermal or non-thermal, in terms of electronic density or temperature for the removal of contaminants (Table 5). Hijosa-Valsero and coworkers (Hijosa-Valsero et al., 2013) have used non-thermal plasma for the degradation of organic micropollutants (atrazine, chlorfenvinphos and 2,4-dibromophenol) from aqueous solutions. Degradation reactions in both dielectric barrier discharge (DBD) devices (planar and coaxial) followed first-order kinetics in distilled water. The highest k value was recorded for 2,4-dibromophenol in the planar reactor, whereas the lowest k value was obtained for atrazine in the coaxial reactor. The planar (and batch) reactor achieved higher removal efficiency than the coaxial reactor (Hijosa-Valsero et al., 2013). DBD plasma showed a significant degradation effects on dimethoate, an organophosphorus pesticide, achieving degradation efficiency of up to 96 % using optimal operating parameters

Table 3
Summary of the photocatalytic process for the removal/degradation of pesticide.

Pesticide	Initial concentration and Matrix-Scale	AOP features	Kinetic data	Reference
Thirty-one pesticides	COD = 1662–1960 mg O ₂ L ⁻¹ , DOC = 513–696 mg C L ⁻¹ , BOD ₅ = 1350–1600 mg O ₂ L ⁻¹ , and nineteen pesticides = 0.02–45 mg L ⁻¹ Present in wastewater	immobilized biomass reactor (IBR)+ solar-driven AOPs; Heterogeneous (TiO ₂ /UV or TiO ₂ /H ₂ O ₂ /UV, both with and without acidification) OR Homogeneous (UV, H ₂ O ₂ /UV, Fe ²⁺ /H ₂ O ₂ /UV or Fe ²⁺ /H ₂ O ₂) phases) pH = Acidic to neutral	Photo-Fenton reaction performed 8.4, 8.7 and 5.1 times higher degradation efficiency than H ₂ O ₂ /UV, TiO ₂ /H ₂ O ₂ /UV-without and with acidification respectively	(Moreira et al., 2012)
Imazalil	25 mg L ⁻¹ using aqueous solution	UV irradiation + TiO ₂ loading (2.5 g L ⁻¹) + Persulfate (~2.5 g L ⁻¹) pH = 3 – 4	Imazalil removal (≤10 min) ~ 100%	(Hazime et al., 2013)
Monocrotophos (MCP) and Chlorpyrifos (CPF)	2 mg L ⁻¹ using aqueous solution	TiO ₂ /UV Catalyst dosage = 1 g L ⁻¹ pH = 5	pseudo-first-order MCP and CPF removal (~60 min) ≥ 98 %	(Amalraj and Pius, 2015)
Chlorpyrifos	2.84 mg L ⁻¹ in aqueous solution	TiO ₂ + synthetic solar radiation using raceway pond reactor (RPR), dosage of TiO ₂ (17.07 mg L ⁻¹) neutral pH	Removal efficiency (55.15 min) = 71.09 ± 1.9 %	(Flint and Suslick, 1991)
Chlorpyrifos	2.74 mg L ⁻¹ in agriculture runoff	TiO ₂ + synthetic solar radiation using RPR, dosage of TiO ₂ (15.72 mg L ⁻¹) neutral pH	Removal efficiency (62.5 min) = 84.01 %	(Amiri et al., 2018)
α-hexachloro cyclohexane (α-HCH), malathion and CPF	50 mg L ⁻¹ in aqueous solution	FeHCF + Sunlight photocatalyst-dose (25 mg), neutral pH	Removal efficiency (12 h) = 84.01 %, malathion (95 %) < CPF (92 %) < α-HCH (90 %)	(Rani and Shanker, 2018)
3,5,6-trichloro-2-pyridinol (TCP)	50 mg L ⁻¹ in aqueous solution	Immobilised TiO ₂ + UVA (315–400 nm) pH = 4.9	TCP removal (120 min irradiation) = 91.0 %, chloride concentration = 26.6 mg L ⁻¹	(Žabar et al., 2016)
Methyl parathion (MP) and parathion (PA)	14 mg mL ⁻¹ in real water/ distilled water	UV (i.e. direct photolysis) and UV–ZnO nanocrystal dosage of ZnO = 160 mg l ⁻¹ pH = 6.7	Removal efficiency UV–ZnO < ZnO < UV Rate of degradation (60 min): MP = 0.874 PA = 0.878	(Sharma et al., 2016)
Acetamiprid	20.55 mg L ⁻¹ in aqueous solution	TiO ₂ /Fe ₃ O ₄ /SiO ₂ nanocomposite (550 mg L ⁻¹) pH = 7.5	Acetamiprid removal (65–73 min) = 76.55%	(Hoffmann et al., 1996)
Diazinon	20 mg L ⁻¹ in aqueous solution	UV (125 W) + ZnO–TiO ₂ (0.5 g L ⁻¹) pH = 7	Pseudo-first order Removal efficiency (120 min) UV/ZnO–TiO ₂ –H ₂ O ₂ (100%) < UV/ZnO–TiO ₂ /O ₂ (99.56%) < UV/ZnO–TiO ₂ –87.26%	(Jonidi-Jafari et al., 2015)
Diazinon	20 mg L ⁻¹ in aqueous solution	UV (125 W) + Cu-doped ZnO (0.5 g L ⁻¹) pH = 7	Pseudo-first order Removal efficiency (120 min), UV/ZnO (58.52 %), UV/Cu-doped ZnO (96.79 %)	(Shirzad-Siboni et al., 2017)

Table 3 (Continued)

Pesticide	Initial concentration and Matrix-Scale	AOP features	Kinetic data	Reference
[-3.6pt] Diazinon	30 mg L ⁻¹ in aqueous solution	Sonophotocatalytic using iron doped TiO ₂ (0.4 g L ⁻¹) pH = 5.5	Removal efficiency (100 min), Ultrasonic/UV/Fe doped TiO ₂ (85%) < UV irradiation (36%) < Iron-doped TiO ₂ (30%) < Ultrasonic (19%)	(Tabasideh et al., 2017)
Diazinon	10 mg L ⁻¹ in aqueous solution	T = 110 °C, visible light (14 W/cm ²) + TiO ₂ /Fe ₂ O ₃ (0.1 g L ⁻¹)	Removal efficiency (45 min), Diazinon = 95.07 %	(Mirmasoomi et al., 2017)
Diazinon	1 mg L ⁻¹ in aqueous solution	TiO ₂ /Fe ₃ O ₄ (0.55 g L ⁻¹) pH = 7.5	Removal efficiency (65 min), Diazinon (94.15 %) COD (99.80 %)	(Suslick, 1990)

Table 4

Summary of the Fenton-type processes for the removal/degradation of pesticides.

Pesticide	Initial concentration and Matrix-Scale	AOP features	Kinetic data	References
Malathion	33 mg L ⁻¹ In synthetic wastewater	Nanofiltration (NF), photo-Fenton Malathion:H ₂ O ₂ =1:100, H ₂ O ₂ : Fe(II)=40:01:00 pH = 3	Removal efficiency(100 min) = ~1.0 %	(Zhang and Pagilla, 2010)
Methyl parathion	aqueous solution	Hydrodynamic cavitation reactor with H ₂ O ₂ and Fenton's reagent pH = 3 H ₂ O ₂ :FeSO ₄ (1:4)	Removal efficiency(100 min) = Methyl parathion ~ 90 % TOC ~76 %	(Patil and Gogate, 2012)
Chlorpyrifos, Cypermethrin, Chlorothalonil	CPF = 805.56 ± 10.0 mg L ⁻¹ , Cypermethrin 105.75 ± 10.0 mg L ⁻¹ , Chlorothalonil 692.08 ± 10.0 mg L ⁻¹ , COD 3350.0 ± 100 mg L ⁻¹ TOC 2960.0 ± 100 mg L ⁻¹ in Pesticide wastewater	UV Fenton pretreatment combined with aerobic sequencing batch reactor (SBR) pH = 3 H ₂ O ₂ : COD (2:1) H ₂ O ₂ : Fe ²⁺ (25:1)	Removal efficiency of UV Fenton (60 min) = COD = 64.8 % TOC = 45.9 % Removal efficiency of UV Fenton-SBR (40 d operation at 12 hr HRT) COD = 96.2 % TOC = 97.4 %	(Affam et al., 2014)
chlorfenvinphos (CFVP)	2.8 × 10 ⁻⁴ M in drinking water networks contaminated with CFVP	Fenton in batch reactor H ₂ O ₂ = 1.5 × 10 ⁻² M Fe ²⁺ = 4.6 × 10 ⁻⁴ M H ₂ O ₂ : FeSO ₄ = 32.6:1 pH = 3 T = 30 °C	CFVP removal (60 min) ~1.0 %	(Oliveira et al., 2014)
Triazophos	20 mg L ⁻¹ in aqueous solution	Hydrodynamic cavitation and Fenton's reagent, Pressure = 5 bar pH = 3 H ₂ O ₂ : FeSO ₄ = 4:1	pseudo first-order rate, Triazophos removal (120 min), hydrodynamic cavitation (50 %) < hydrodynamic cavitation and Fenton (~83 %) < hydrodynamic cavitation + ozonation (90 min) (Triazophos, ~100 %, TOC = 96 %)	(Gogate and Patil, 2015)
Diazinon	50 mg L ⁻¹ in aqueous solution	Sono-Fenton and Sono-Fenton-like systems ultrasonic probe (frequency 20 kHz) Fe ²⁺ = 20 mg L ⁻¹ , H ₂ O ₂ = 150 mg L ⁻¹ , T = 25 °C pH = 3	pseudo first-order rate Diazinon removal (60 min), ultrasound, ultrasound + Fe ²⁺ and ultrasound + H ₂ O ₂ (25.7 %), Ultrasound + Fenton Diazinon = 98.3 % mineralization = 29.9 %	(Wang and Shih, 2015)

Table 5
Summary of the plasma water technology for the removal/degradation of pesticides.

Pesticide	Initial concentration and Matrix-Scale	AOP features	Kinetics	References
Atrazine, chlorfenvinfos, 2,4-dibromophenol, and lindane	Atrazine(5 mg L ⁻¹), Chlorfenvinfos (5 mg L ⁻¹), 2,4-dibromophenol, (1 mg L ⁻¹), Lindane (1 mg L ⁻¹), in Distilled water	Non-thermal plasma, DBD plasma reactor	first-order degradation kinetics, Batch reactor removal efficiency (5 min), 2,4-dibromophenol (98 %) < Chlorfenvinfos (94.3 %) < Atrazine (93.2 %) < Lindane (86.6 %)	(Hijosa-Valsero et al., 2013)
Dimethoate	20 mgL ⁻¹ in aqueous solution	DBD plasma, power = 85 W, air-gap distance = 5 mm	removal efficiency (7 min), < 96 %	(Hu et al., 2013)
Dichlorvos, malathion, endosulfan	Dichlorvos (850 ppb), malathion (1320 ppb), endosulfan (350 ppb) in aqueous solution	DBD plasma, 80 kV	first order kinetics, Removal efficiency (8 min), dichlorvos (78.98 ± 0.81 %) <, malathion (69.62 ± 0.14 %) <, endosulfan (57.71 ± 0.58 %)	(Sarangapani et al., 2016)

Table 6
Summary of the gamma irradiation technology for the removal/degradation of pesticides.

Pesticide	Initial concentration and Matrix-Scale	AOP features	Kinetics	References
Chlorpyrifos	5–20 mgL ⁻¹ in Distilled water / lake water	Cobalt-60 gamma irradiation and sunlight T = 30 °C pH = Neutral	CPF removal efficiency; 12 days using sunlight (CPF = 20 mgL ⁻¹), lake water (51.95 %) < distilled water (39.5 %), 12 kGy using gamma (CPF = 5 mgL ⁻¹) (~100%)	(Hossain et al., 2013)
Chlorpyrifos	500 µg L ⁻¹ in aqueous solution	Cobalt-60 gamma irradiation, dose rate (300 Gy h ⁻¹)	pseudo-first order, CPF removal efficiency (absorbed dose of 575 Gy), ~100 %	(Ismail et al., 2013)

(Table 5)(Hu et al., 2013). Degradation of dichlorvos by DBD plasma was higher than that of malathion and endosulfan (Sarangapani et al., 2016).

4.2.6. Gamma irradiation technology

The irradiation of water molecules by gamma rays produces mainly the OH[•] radicals in addition to solvated electrons (e_{aq}⁻), hydrogen radicals (H[•]), superoxide radical anions (O₂^{-•}), hydroperoxyl radical (HO₂[•]), hydrogen peroxide (H₂O₂) and hydrogen (H₂). Gamma radiation has proven more effective for the treatment of aqueous contaminants than other methods because hydroxyl radicals can easily form. In addition, gamma irradiation can remove organophosphorus pesticides at low concentrations (Khedr et al., 2019). Hossain and coworkers (Hossain et al., 2013) compared gamma irradiation and natural sunlight for the oxidation of chlorpyrifos in aqueous solutions. According to their findings, Table 6, gamma-ray irradiation could be used in combination with conventional sunlight methods to degrade environmental samples contaminated with the organophosphate pesticide such as chlorpyrifos. Although the degradation of chlorpyrifos using sunlight took a long time (12 days), it was more efficient compared to gamma irradiation with high energy consumption. In a similar study, Ismail and coworkers (Ismail et al., 2013) used gamma irradiation to study the removal efficiency of chlorpyrifos in aqueous solution by 60Co c-rays on a laboratory scale. These authors confirmed that the application of radiation technology could be an effective method to destroy water chlorpyrifos (Table 6). In fact, the concentration of chlorpyrifos was reduced from 500 µg L⁻¹ to

below the detection limit (5 µg L⁻¹) of the analytical method in aerated solutions at an absorbed dose of 575 Gy.

4.2.7. Sulfate-based catalyst

Persulfate activation can be initiated by thermal or chemical means to form sulfate radicals, SO₄^{•-}. The sulfate radical is a stronger oxidant (E° = 2.6 V) than the persulfate anion. Molybdenum disulfide (MoS₂) is a catalyst that, due to its special features such as large surface area and catalytically active sites, attracts great attention in the AOPs. In addition, MoS₂ has a narrow band gap that improves absorption of visible light and electron-hole pairs upon light excitation, which makes it a candidate for photocatalytic applications under visible light. Aimer et al., (Aimer et al., 2019) studied the removal efficiency of dimethoate (DIM), an organophosphorus pesticide, by both sulfate radicals of heat-activation of persulfate (PS) and hydroxyl radicals produced by electro-oxidation using a BDD anode, as shown in Table 7. The removal efficiency using PS was higher than that of the DIM electro-oxidation process. However, the BOD₅/COD ratio during the electrochemical oxidation of DIM via hydroxyl radicals showed that it was possible to render the solution biodegradable without reaching a complete mineralization. Fenitrothion is a common organophosphorus pesticide that was completely removed by PS activation by zero-valent iron powder (Fe⁰). Fe⁰ enhanced the degradation process and reduced the energy barrier during the degradation of pollutants using PS (Liu et al., 2019) (Table 7).

Table 7

Summary of sulfate-based catalyst for the removal/degradation of pesticides.

Pesticide	Initial concentration and Matrix-Scale	AOP features	Kinetic data	References
Dimethoate (DIM)	0.1 mM in Synthetic wastewater	heat-activation PS / electro oxidation T = 60 °C	DIM removal efficiency: (90 min, [DIM]/[PS] = 1/50) ~100 % electro oxidation(120 min), ~100 %	(Aimer et al., 2019)
Fenitrothion (FNT)	10 mg L ⁻¹ and 20 µg L ⁻¹ in Ultra-pure water	Persulfate (PS) activation by zero-valent iron powder (Fe0) pH = 3.0, T = 35 °C Molar ratio FNT:PS = 1:500 Fe0:PS = 1:1.5	Pseudo-first-order Fenitrothion (FNT = 10 mg L ⁻¹ , 45 min) ~100% (FNT = 20 µg L ⁻¹ , 12 h) ~ 96 ± 0%	(Liu et al., 2019)

Table 8

Summary of the sonolysis process for the removal/degradation of pesticides.

Pesticide	Initial concentration and Matrix-Scale	AOP features	Kinetic data	References
Azinphos-methyl / chlorpyrifos	1 mg L ⁻¹ in aqueous solution	Ultrasound treatment, pH = 9, T = 25.0 ± 1.0 °C frequency = 130 kHz, electric power = 500 W,	Removal efficiency (20 min), CPF = 98.96 % Azinphos-methyl = 78.50 %	(Agarwal et al., 2016)
Diazinon	50 mg L ⁻¹ in aqueous solution	Ultrasound facilitated by Fenton's and Fenton-like reagents, pH = 3 T = 25 °C Fe ²⁺ = 20 mg L ⁻¹ H ₂ O ₂ = 150 mg L ⁻¹ ,	Diazinon removal efficiency (60 min), ~ 98 % Mineralization ~ 30 %	(Wang and Shih, 2016)

4.2.8. Sonolysis

In the sonochemical process the main mechanism for the destruction of the organic pollutants is the formation of hydroxyl radicals, oxygen atoms and hydrogen atoms as a result of water pyrolysis. Ultrasonic pressure waves, made by ultrasonic irradiation (20–500 kHz) of aqueous solutions, result in the formation of vapor bubbles. These bubbles collapse violently after reaching a critical resonance size and create transient high temperatures (>5000 K) (Flint and Suslick, 1991), high pressures (>1000 bar) and highly reactive radicals (Suslick, 1990). Thermal decomposition and numerous radical reactions take place during the degradation of water contaminants (Hoffmann et al., 1996; Yaghmaei et al., 2016). The critical parameters in sonolysis include pH, initial pesticide concentration, frequency, electric power and treatment time. Table 8 shows the results of two studies related to pesticide degradation by this process. Factors influencing the sonolytic degradation of azinphos-methyl and chlorpyrifos (Agarwal et al., 2016) and the degradation of insecticide diazinon by the Fenton and Fenton-like processes combined with ultrasound (Wang and Shih, 2016) were investigated. It was reported (Agarwal et al., 2016) that the initial concentration of pesticides had a strong effect on the removal efficiency of azinphos-methyl and chlorpyrifos. It was also observed that the degradation of pesticides decreased with the increase of the initial concentration of pollutant, while the pH variation did not affect the degradation of pesticides (Agarwal et al., 2016). The use of ultrasound requires high-energy consumption, which results in a very low electrical efficiency compared to other AOPs technologies. Because of this, the hybrid technology, which combines ultrasound with other AOPs, can provide more efficiency and additional benefits. Wang and Shih (Wang and Shih, 2016) considered the effects of oxidants, such as persulphate ions (S₂O₈²⁻) and hydrogen peroxide (H₂O₂), transition metal (including Co²⁺,

Ag⁺ and Fe²⁺), the initial concentrations of iron/H₂O₂ in the Fenton process and the temperature on the degradation of diazinon. The diazinon was effectively degraded, and enhanced the toxicity reduction, by the combination of the ultrasonic irradiation with the Fenton and Fenton-like processes (Wang and Shih, 2016).

4.2.9. Ozone based AOPs

Ozone is a strong oxidant itself with an oxidation potential of 2.07 V. Ozone can react in aqueous solution either directly with target substrates (in ionized and dissociated form of organic compounds, instead of the neutral form) or indirectly through reactions with its free radical decomposition products (Kang and Hoffmann, 1998). The ozonation process is often applied simultaneously with other AOPs, such as UV radiation, hydrogen peroxide, ultrasound, activated carbon and catalysts to (a) improve the generation of •OH radicals, (b) enhance the oxidation rate of contaminants, (c) lower ozone consumption and, (d) convert contaminants into less toxic species which improves the biodegradability of the by-products. The ozonation process was applied to degrade omethoate (OMT), acetamiprid (ACMP) and dichlorvos (DDVP) pesticides (Table 9). The combination of O₃ with Fe(III)-loaded activated carbon(Fe@AC), for the effective generation of hydroxyl radicals, accelerated the degradation of OMT and its dimethoate byproduct, as compared to ozonation alone and O₃/AC (Ling et al., 2011). In the oxidation of ACMP by ozonation, second-order kinetic constants were recorded for the ACMP reactions with molecular ozone (0.25 M⁻¹ s⁻¹) and hydroxyl radicals (2.1 × 10⁹ M⁻¹ s⁻¹). Thus, such values of the kinetic constants showed the resistance of the pesticide structure alongside ozone. Although the ozone dose of 5.5 mg L⁻¹ was able to completely remove the ACMP with an initial concentration of 10 µM, its main intermediate products required higher doses of ozone (Cruz-Alcalde et al., 2017). Cruz-Alcalde and

Table 9
Summary of the ozone based AOPs process for the removal/degradation of pesticides.

Pesticide	Initial concentration and Matrix-Scale	AOP features	Kinetic data	References
Omethoate (OMT)	1.0 mg L ⁻¹ in aqueous solution	Fe(III)-loaded activated carbon + Ozone, pH = 7–8, T = 20 ± 2 °C, catalyst dose = 20 mg L ⁻¹ , O ₃ = 1.0 mg L ⁻¹	OMT removal efficiency (30 min), ozonation alone (37.6 %) < O ₃ /AC (58.0 %) < O ₃ /5 %-Fe@AC (82.4 %)	(Ling et al., 2011)
Acetamiprid (ACMP)	10 µM in aqueous solution	Ozonation; Ozone dosage (5.5 mg L ⁻¹), pH = 7, T = 20 ± 2 °C	Second order rate constants, ACMP removal efficiency ~ 100 %	(Cruz-Alcalde et al., 2017)
Dichlorvos (DDVP)	20 µM in aqueous solution	Ozonation; Ozone dosage (5.5 mg L ⁻¹), pH = 7, T = 20 ± 2 °C	Second order rate constants, ASMP removal efficiency ~ 100 %	(Cruz-Alcalde et al., 2018)

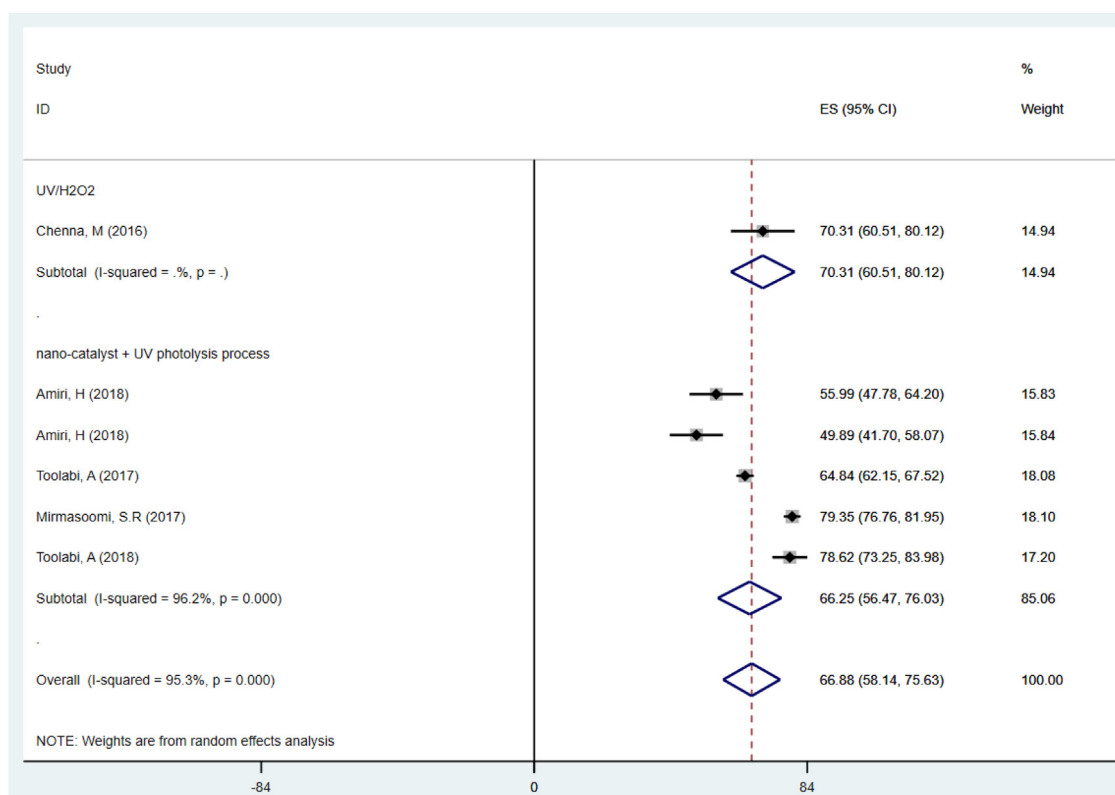


Fig. 5. The forest plot of the organophosphorus pesticide removal by type of AOP subgroup.

Esplugas reported the removal of DDVP with the ozonation process. These authors observed that DDVP exhibited moderate reactivity with molecular ozone, but indirect oxidation by $\cdot\text{OH}$ guaranteed the DDVP and the associated toxicity abatement. It was reported that second-order rate constants of the DDVP reactions with O_3 and $\cdot\text{OH}$ were determined as 590 and $2.2 \cdot 10^9 \text{ M}^{-1} \text{ s}^{-1}$, respectively in neutral pH (Cruz-Alcalde et al., 2018).

4.2.10. Other AOPs

Bustos and coworkers have evaluated the potential photodegradation of dichlorvos (DDVP) under simulated sunlight and UV-254 irradiation (Table 10) (Bustos et al., 2019). In addition, the influence on this process of dissolved oxygen (DO) and dissolved organic matter (DOM) –especially the humic fraction of DOM– on the degra-

dation of DDVP was carefully considered. The presence of DO in ultrapure water significantly improved the photodegradation of DDVP, which proved the importance of the role of oxygen in the degradation of pesticides in natural waters under solar radiation. Although UVC shows more removal efficiency compare to sunlight, more energy and photons flux are necessary to achieve the desired condition (Bustos et al., 2019).

4.3. Meta-analysis

The meta-analysis was performed considering six studies. All studies, except one, used photocatalytic process as AOP. Four types of pesticide were used in these studies, such as chlorpyrifos, reported in two studies (Amiri et al., 2018; Naddafi et al., 2018),

Table 10
Summary of different AOPs processes for the removal/degradation of pesticides.

Pesticide	Initial concentration and Matrix-Scale	AOP features	Kinetic data	References
Dichlorvos (DDVP)	4.5 μ M in Deionized water/ Wastewater treatment plant effluent	Photodegradation (sunlight irradiation, UV-C) pH = 3.0 ± 0.2 , 7.0 ± 0.2	DDVP removal efficiency: pH = 3 / presence of DO > pH = 7 / absent of DO, UVC > Sunlight	(Bustos et al., 2019)

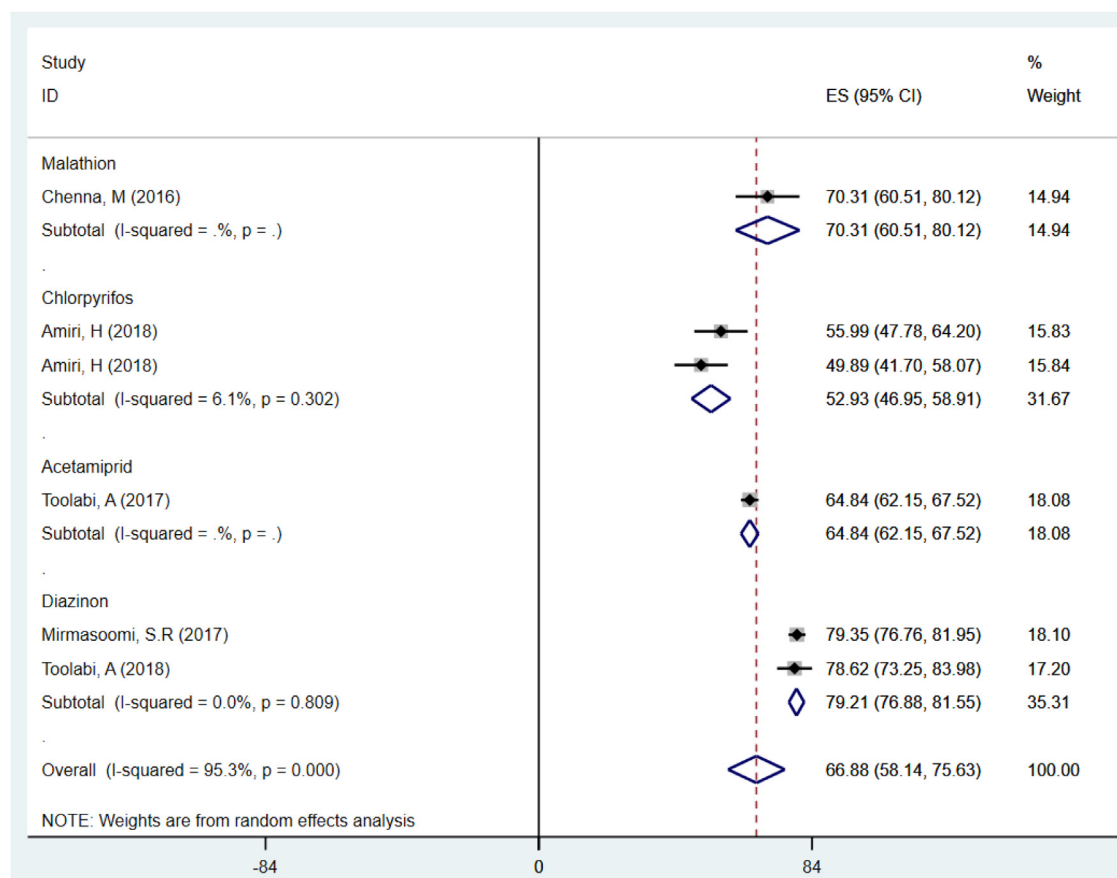


Fig. 6. The forest plot of the organophosphorus pesticide removal by type of pesticide subgroup.

diazinon, reported also in two studies (Mirmasoomi et al., 2017; Toolabi et al., 2019) and, in one study, malathion and acetamiprid were reported (Toolabi et al., 2017).

The pooled mean percentage of AOP for pesticide degradation was 66.8 (95 %CI: 58.1–75.6). After excluding the only study in which UV/H₂O₂ was used, the overall average percentage of AOP for pesticide degradation using the photocatalytic process was 66.2 (95 % CI: 76.0–85.0), as shown in Fig. 5. We also conducted a subgroup analysis based on the type of pesticide used. In the studies, which used the photocatalytic process as AOP, the highest removal efficiency was obtained for diazinon with an average percentage of 79.2 (95 % CI: 76.8–81.5). The pooled removal efficiency for chlorpyrifos was 52.93 (95 %CI: 46.9–58.9), as depicted in Fig. 6.

5. Conclusions

This study constitutes the first systematic review exclusively focused on the treatment of organophosphorus pesticides by several AOPs. The available literature reviewed here has shown a growing interest in recent years in AOPs applications for the

removal of pesticides from aqueous media. It has been clearly demonstrated that most of the pesticides reviewed in this document are substantially reactive and readily degradable by various AOPs. The most studied pesticides are chlorpyrifos and diazinon, since they have been identified as the main ones available and usable in farmlands. In addition, the results of the meta-analysis of some papers confirm that the common of the AOPs tested showed a remarkable efficiency in the destruction of the target compounds by 70 %. Among the AOPs, photocatalytic and Fenton processes appear to be the most common technologies for the degradation of organophosphorus pesticides.

Declaration of Competing Interest

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version. This manuscript has not been submitted to, nor is under review at, another journal or other publishing venue. The authors have no affiliation with any organization with a

direct or indirect financial interest in the subject matter discussed in the manuscript.

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References

- Affam, A.C., Chaudhuri, M., Kutty, S.R.M., Muda, K., 2014. UV Fenton and sequencing batch reactor treatment of chlorpyrifos, cypermethrin and chlorothalonil pesticide wastewater. *Int. Biodeterior. Biodegradation* 93, 195–201.
- Agarwal, S., Tyagi, I., Gupta, V.K., Dehghani, M.H., Bagheri, A., Yetilmezsoy, K., Amrane, A., Heibati, B., Rodriguez-Couto, S., 2016. Degradation of azinphos-methyl and chlorpyrifos from aqueous solutions by ultrasound treatment. *J. Mol. Liq.* 221, 1237–1242.
- Aimer, Y., Benali, O., Groenen Serrano, K., 2019. Study of the degradation of an organophosphorus pesticide using electrogenerated hydroxyl radicals or heat-activated persulfate. *Sep. Purif. Technol.* 208, 27–33.
- Amalraj, A., Pius, A., 2015. Photocatalytic degradation of monocrotophos and chlorpyrifos in aqueous solution using TiO₂ under UV radiation. *J. Water Process. Eng.* 7, 94–101.
- Amiri, H., Nabizadeh, R., Silva Martinez, S., Jamaledin Shahtaheri, S., Yaghmaeian, K., Badiei, A., Nazmara, S., Naddafi, K., 2018. Response surface methodology modeling to improve degradation of Chlorpyrifos in agriculture runoff using TiO₂ solar photocatalytic in a raceway pond reactor. *Ecotoxicol. Environ. Saf.* 147, 919–925.
- Ballantyne, B., Marrs, T.C., 2017. *Clinical and Experimental Toxicology of Organophosphates and Carbamates*. Elsevier.
- Bustos, N., Cruz-Alcalde, A., Iriel, A., Fernández Cirelli, A., Sans, C., 2019. Sunlight and UVC-254 irradiation induced photodegradation of organophosphorus pesticide dichlorvos in aqueous matrices. *Sci. Total Environ.* 649, 592–600.
- Chenna, M., Messaoudi, K., Drouiche, N., Lounici, H., 2016. Study and modeling of the organophosphorus compound degradation by photolysis of hydrogen peroxide in aqueous media by using experimental response surface design. *J. Ind. Eng. Chem.* 33, 307–315.
- Cruz-Alcalde, A., Sans, C., Esplugas, S., 2017. Priority pesticides abatement by advanced water technologies: the case of acetamiprid removal by ozonation. *Sci. Total Environ.* 599–600, 1454–1461.
- Cruz-Alcalde, A., Sans, C., Esplugas, S., 2018. Priority pesticide dichlorvos removal from water by ozonation process: reactivity, transformation products and associated toxicity. *Sep. Purif. Technol.* 192, 123–129.
- Dominguez, C.M., Oturan, N., Romero, A., Santos, A., Oturan, M.A., 2018. Optimization of electro-Fenton process for effective degradation of organochlorine pesticide lindane. *Catal. Today* 313, 196–202.
- Dzyadevych, S.V., Chovelon, J.-M., 2002. A comparative photodegradation studies of methyl parathion by using Lumistox test and conductometric biosensor technique. *Mater. Sci. Eng. C* 21, 55–60.
- Fenoll, J., Hellín, P., Martínez, C.M., Flores, P., Navarro, S., 2012. Semiconductor oxides-sensitized photodegradation of fenamiphos in leaching water under natural sunlight. *Appl. Catal. B: Environ.* 115–116, 31–37.
- Flint, E.B., Suslick, K.S., 1991. The temperature of cavitation. *Science (New York, N.Y.)* 253, 1397–1399.
- Gogate, P.R., Patil, P.N., 2015. Combined treatment technology based on synergism between hydrodynamic cavitation and advanced oxidation processes. *Ultrason. Sonochem.* 25, 60–69.
- Guivarch, E., Oturan, N., Oturan, M.A., 2003. Removal of organophosphorus pesticides from water by electrogenerated Fenton's reagent. *Environ. Chem. Lett.* 1, 165–168.
- Hazime, R., Nguyen, Q.H., Ferronato, C., Huynh, T.K.X., Jaber, F., Chovelon, J.M., 2013. Optimization of imazalil removal in the system UV/TiO₂/K₂S₂O₈ using a response surface methodology (RSM). *Appl. Catal. B: Environ.* 132–133, 519–526.
- Herrmann, J.M., Guillard, C., Arguello, M., Agüera, A., Tejedor, A., Piedra, L., Fernandez-Alba, A., 1999. Photocatalytic degradation of pesticide pirimiphos-methyl: determination of the reaction pathway and identification of intermediate products by various analytical methods. *Catal. Today* 54, 353–367.
- Hijosa-Valsero, M., Molina, R., Schikora, H., Müller, M., Bayona, J.M., 2013. Removal of priority pollutants from water by means of dielectric barrier discharge atmospheric plasma. *J. Hazard. Mater.* 262, 664–673.
- Hoffmann, M.R., Hua, I., Höchemer, R., 1996. Application of ultrasonic irradiation for the degradation of chemical contaminants in water. *Ultrason. Sonochem.* 3, S163–S172.
- Hossain, M.S., Fakhruddin, A.N.M., Chowdhury, M.A.Z., Alam, M.K., 2013. Degradation of chlorpyrifos, an organophosphorus insecticide in aqueous solution with gamma irradiation and natural sunlight. *J. Environ. Chem. Eng.* 1, 270–274.
- Hu, Y., Bai, Y., Li, X., Chen, J., 2013. Application of dielectric barrier discharge plasma for degradation and pathways of dimethoate in aqueous solution. *Sep. Purif. Technol.* 120, 191–197.
- Ismail, M., Khan, H.M., Sayed, M., Cooper, W.J., 2013. Advanced oxidation for the treatment of chlorpyrifos in aqueous solution. *Chemosphere* 93, 645–651.
- Jaafarzadeh, N., Ahmadi, M., Silva Martínez, S., Amiri, H., 2014. Removal of As (III) and As (V) from aqueous solution using modified solid waste vegetable oil industry as a natural adsorbent. *Environmental Engineering & Management Journal (EEMJ)*, 13.
- Jonidi-Jafari, A., Shirzad-Siboni, M., Yang, J.-K., Naimi-Joubani, M., Farrokhi, M., 2015. Photocatalytic degradation of diazinon with illuminated ZnO–TiO₂ composite. *J. Taiwan Inst. Chem. Eng.* 50, 100–107.
- Kang, J.-W., Hoffmann, M.R., 1998. Kinetics and mechanism of the sonolytic destruction of methyl tert-butyl ether by ultrasonic irradiation in the presence of ozone. *Environ. Sci. Technol.* 32, 3194–3199.
- Kang, J., Zettl, V., Ward, N., 1995. The organophosphate pesticides. *J. Nutr. Environ. Med.* 5, 325–339.
- Katsikantami, I., Colosio, C., Alegakis, A., Tzatzarakis, M.N., Vakonaki, E., Rizos, A.K., Sarigiannis, D.A., Tsatsakis, A.M., 2019. Estimation of daily intake and risk assessment of organophosphorus pesticides based on biomonitoring data – the internal exposure approach. *Food Chem. Toxicol.* 123, 57–71.
- Kazemi, M., Tahmasbi, A.M., Valizadeh, R., Naserian, A.A., Soni, A., 2012. Organophosphate pesticides: a general review. *Agric. Sci. Res. J.* 2, 512–522.
- Khedr, T., Hammad, A.A., Elmarsafy, A.M., Halawa, E., Soliman, M., 2019. Degradation of some organophosphorus pesticides in aqueous solution by gamma irradiation. *J. Hazard. Mater.* 373, 23–28.
- Kukurina, O., Elmesova, Z., Syskina, A., 2014. Mineralization of organophosphorous pesticides by electro-generated oxidants. *Procedia Chem.* 10, 209–216.
- Lebik-Elhadi, H., Frontistis, Z., Ait-Amar, H., Amrani, S., Mantzavinos, D., 2018. Electrochemical oxidation of pesticide thiamethoxam on boron doped diamond anode: role of operating parameters and matrix effect. *Process. Saf. Environ. Prot.* 116, 535–541.
- Li, W., Liu, Y., Duan, J., van Leeuwen, J., Saint, C.P., 2015. UV and UV/H₂O₂ treatment of diazinon and its influence on disinfection byproduct formation following chlorination. *Chem. Eng. J.* 274, 39–49.
- Ling, W., Qiang, Z., Shi, Y., Zhang, T., Dong, B., 2011. Fe(III)-loaded activated carbon as catalyst to improve omethoate degradation by ozone in water. *J. Mol. Catal. A Chem.* 342–343, 23–29.
- Liu, H., Yao, J., Wang, L., Wang, X., Qu, R., Wang, Z., 2019. Effective degradation of fenitrothion by zero-valent iron powder (Fe⁰) activated persulfate in aqueous solution: kinetic study and product identification. *Chem. Eng. J.* 358, 1479–1488.
- Mahvi, A.H., Malakootian, M., Heidari, M.R., 2011. Comparison of polyaluminum silicate chloride and electrocoagulation process, in natural organic matter removal from surface water in Ghochan. *Iran. Journal of Water Chemistry and Technology* 33, 377–385.
- Malakootian, M., Moridi, A., 2017. Efficiency of electro-Fenton process in removing Acid Red 18 dye from aqueous solutions. *Process. Saf. Environ. Prot.* 111, 138–147.
- Mirmasoomi, S.R., Mehdipour Ghazi, M., Galedari, M., 2017. Photocatalytic degradation of diazinon under visible light using TiO₂/Fe₂O₃ nanocomposite synthesized by ultrasonic-assisted impregnation method. *Sep. Purif. Technol.* 175, 418–427.
- Moreira, F.C., Vilar, V.J.P., Ferreira, A.C.C., dos Santos, F.R.A., Dezotti, M., Sousa, M.A., Gonçalves, C., Boaventura, R.A.R., Alpendurada, M.F., 2012. Treatment of a pesticide-containing wastewater using combined biological and solar-driven AOPs at pilot scale. *Chem. Eng. J.* 209, 429–441.
- Naddafi, K., Nabizadeh, R., Silva-Martinez, S., Shahtaheri, S.J., Yaghmaeian, K., Badiei, A., Amiri, H., 2018. Modeling of Chlorpyrifos degradation by TiO₂ photo catalysis under visible light using response surface methodology. *Desalin. Water Treat.* 106, 220–225.
- Oliveira, C., Alves, A., Madeira, L.M., 2014. Treatment of water networks (waters and deposits) contaminated with chlorfenvinphos by oxidation with Fenton's reagent. *Chem. Eng. J.* 241, 190–199.
- Patil, P.N., Gogate, P.R., 2012. Degradation of methyl parathion using hydrodynamic cavitation: effect of operating parameters and intensification using additives. *Sep. Purif. Technol.* 95, 172–179.
- Priya, D.N., Modak, J.M., Trebše, P., Žabar, R., Raichur, A.M., 2011. Photocatalytic degradation of dimethoate using LbL fabricated TiO₂/polymer hybrid films. *J. Hazard. Mater.* 195, 214–222.
- Rani, M., Shanker, U., 2018. Effective adsorption and enhanced degradation of various pesticides from aqueous solution by Prussian blue nanorods. *J. Environ. Chem. Eng.* 6, 1512–1521.
- Ren, Q., Yin, C., Chen, Z., Cheng, M., Ren, Y., Xie, X., Li, Y., Zhao, X., Xu, L., Yang, H., Li, W., 2019. Efficient sonoelectrochemical decomposition of chlorpyrifos in aqueous solution. *Microchem. J.* 145, 146–153.
- Samet, Y., Agengui, L., Abdelhédi, R., 2010. Anodic oxidation of chlorpyrifos in aqueous solution at lead dioxide electrodes. *J. Electroanal. Chem.* 650, 152–158.
- Sarangapani, C., Misra, N.N., Milosavljevic, V., Bourke, P., O'Regan, F., Cullen, P.J., 2016. Pesticide degradation in water using atmospheric air cold plasma. *J. Water Process. Eng.* 9, 225–232.
- Sharma, A.K., Tiwari, R.K., Gaur, M.S., 2016. Nanophotocatalytic UV degradation system for organophosphorus pesticides in water samples and analysis by Kubista model. *Arab. J. Chem.* 9, S1755–S1764.
- Shirzad-Siboni, M., Jonidi-Jafari, A., Farzadkia, M., Esrafil, A., Gholami, M., 2017. Enhancement of photocatalytic activity of Cu-doped ZnO nanorods for the degradation of an insecticide: kinetics and reaction pathways. *J. Environ. Manage.* 186, 1–11.
- Sunil Paul, M.M., Aravind, U.K., Pramod, G., Aravindakumar, C.T., 2013. Oxidative degradation of fenvalfothion by hydroxyl radical in aqueous medium. *Chemosphere* 91, 295–301.
- Suslick, K.S., 1990. *Sonochemistry*. Science (New York, N.Y.) 247, 1439–1445.

- Tabasideh, S., Maleki, A., Shahmoradi, B., Ghahremani, E., McKay, G., 2017. Sonophotocatalytic degradation of diazinon in aqueous solution using iron-doped TiO₂ nanoparticles. *Sep. Purif. Technol.* 189, 186–192.
- Toolabi, A., Malakootian, M., Ghaneian, M.T., Esrafil, A., Ehrampoush, M.H., Askar-Shahi, M., Tabatabaei, M., Khatami, M., 2019. Optimizing the photocatalytic process of removing diazinon pesticide from aqueous solutions and effluent toxicity assessment via a response surface methodology approach. *Rend. Lincei Sci. Fis. Nat.* 30, 155–165.
- Toolabi, A., Malakootian, M., Ghaneian, M.T., Esrafil, A., Ehrampoush, M.H., Tabatabaei, M., AskarShahi, M., 2017. Optimization of photochemical decomposition acetamiprid pesticide from aqueous solutions and effluent toxicity assessment by *Pseudomonas aeruginosa* BCRC using response surface methodology. *AMB Express* 7, 159.
- Utzig, L.M., Lima, R.M., Gomes, M.F., Ramsdorf, W.A., Martins, L.R.R., Liz, M.V., Freitas, A.M., 2019. Ecotoxicity response of chlorpyrifos in *Aedes aegypti* larvae and *Lactuca sativa* seeds after UV/H₂O₂ and UVC oxidation. *Ecotoxicol. Environ. Saf.* 169, 449–456.
- Wang, C.-K., Shih, Y.-H., 2016. Facilitated ultrasonic irradiation in the degradation of diazinon insecticide. *Sustain. Environ. Res.* 26, 110–116.
- Wang, C., Shih, Y., 2015. Degradation and detoxification of diazinon by sono-Fenton and sono-Fenton-like processes. *Sep. Purif. Technol.* 140, 6–12.
- Wu, C., Linden, K.G., 2010. Phototransformation of selected organophosphorus pesticides: roles of hydroxyl and carbonate radicals. *Water Res.* 44, 3585–3594.
- Wu, R.-J., Chen, C.-C., Lu, C.-S., Hsu, P.-Y., Chen, M.-H., 2010. Phorate degradation by TiO₂ photocatalysis: parameter and reaction pathway investigations. *Desalination* 250, 869–875.
- Yaghmaeian, K., Silva Martinez, S., Hoseini, M., Amiri, H., 2016. Optimization of As(III) removal in hard water by electrocoagulation using central composite design with response surface methodology. *Desalin. Water Treat.* 57, 27827–27833.
- Žabar, R., Dolenc, D., Jerman, T., Franko, M., Trebše, P., 2011. Photolytic and photocatalytic degradation of 6-chloronicotinic acid. *Chemosphere* 85, 861–868.
- Žabar, R., Sarakha, M., Lebedev, A.T., Polyakova, O.V., Trebše, P., 2016. Photochemical fate and photocatalysis of 3,5,6-trichloro-2-pyridinol, degradation product of chlorpyrifos. *Chemosphere* 144, 615–620.
- Zhang, Y., Pagilla, K., 2010. Treatment of malathion pesticide wastewater with nanofiltration and photo-Fenton oxidation. *Desalination* 263, 36–44.